

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	3	4	2	1	4	3	3	2	3	2	1	2	4	3	2	1	4	2	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	3	1	2	3	2	4	1	2	1	3	3	1	1	1	2	4	4	3	3
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	2	1	2	3	1	2	2	2	2	3	2	4	3	4	3	1	4	3	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
A.	2	4	2	3	1	1	1	4	4	2	3	2	3	2	4	2	2	4	4	3
Q.	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	4	4	3	2	2	3	1	1	1	3	1	1	2	2	3	2	1	1	2	3
Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	2	1	1	3	2	1	2	3	2	1	1	3	3	3	4	2	2	2	2
Q.	121	122	123	124	125	126	127	128	129	130										
A.	3	1	2	1	4	1	3	3	1	3										

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HINT – SHEET

1. **Ans (1)**

$$F = Kx$$

$$U = - \int f dx$$

$$= - \int kx dx$$

$$= - \frac{1}{2} kx^2$$

Work energy theorem

2. **Ans (3)**

$$|\vec{A} \times \vec{B}| = \vec{A} \cdot \vec{B}$$

$$AB \sin \theta = AB \cos \theta$$

$$\tan \theta = 1$$

$$\theta = 45^\circ$$

$$|\vec{A} \times \vec{B}| = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$= \sqrt{A^2 + B^2 + 2AB \cos 45} = \sqrt{A^2 + B^2 + \sqrt{2}AB}$$

3. **Ans (4)**

$$B = [L]$$

$$U = \frac{A\sqrt{x}}{x+B}$$

$$ML^2T^{-2} = \frac{AL^{1/2}}{L}$$

$$A = [ML^{5/2}T^{-2}]$$

$$AB = [ML^{7/2}T^{-2}]$$

4. **Ans (2)**

$$E_1 = \frac{1}{2}Kx^2 \Rightarrow x = \sqrt{\frac{2E_1}{K}}$$

$$E_2 = \frac{1}{2}Ky^2 \Rightarrow y = \sqrt{\frac{2E_2}{K}}$$

$$\text{and } E = \frac{1}{2}K(x+y)^2 \Rightarrow x+y = \sqrt{\frac{2E}{K}}$$

$$\Rightarrow \sqrt{\frac{2E_1}{K}} + \sqrt{\frac{2E_2}{K}} = \sqrt{\frac{2E}{K}} \Rightarrow \sqrt{E_1} + \sqrt{E_2} = \sqrt{E}$$

5. **Ans (1)**

Applying conservation of angular momentum

$$L_i = L_f$$

$$0 = I\omega - mvr$$

$$200 \times \omega = 50 \times 2 \times 1 \Rightarrow \omega = \frac{1}{2} \text{ rad/sec}$$

$$\omega_{\text{platform}} = \frac{1}{2}, \quad \omega_{\text{man}} = \frac{v}{r} = \frac{1}{2}$$

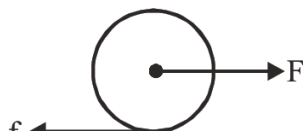
$$\omega_{\text{rel}} = \omega_{\text{platform}} + \omega_{\text{man}} = 1$$

$$\Delta\theta_{\text{rel}} = 2\pi, t = \frac{(\Delta\theta)_{\text{rel}}}{\omega_{\text{rel}}}$$

6. **Ans (4)**

$$F - f = ma \text{ and } f = \mu mg$$

$$f \times R = I\alpha$$



$$f \times R = \frac{2}{5}mR^2 \times \frac{a}{R} \Rightarrow f = \frac{2}{5}ma$$

7. **Ans (3)**

$$f_k = \mu N = 0.04 \times 20g = 8 \text{ Newton}$$

$$a = \frac{60 - 8}{26} = 2 \text{ m/s}^2$$

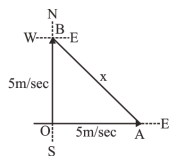
8. Ans (3)

$$a = g \tan \theta$$

$$\Rightarrow a = 10 \times \tan 37^\circ = 10 \times \frac{3}{4} = 10 \times 0.75 = 7.5 \text{ m/s}^2$$

9. Ans (2)

Let OA and OB represents the respective velocities in the east and north directions. Then, AB = x represent the change in velocity.



From figure, $x^2 = 5^2 + 5^2 = 2 \times 5^2$

or $x = \sqrt{2} \times 5 \text{ m/s}$

Now acceleration, $a = \frac{x}{t}$

$$\therefore a = \frac{\sqrt{2} \times 5}{10} = \frac{1}{\sqrt{2}} \text{ m/sec}^2 \quad (\text{towards N-W}).$$

10. Ans (3)

Initial velocity = $u \cos \theta \hat{i} + u \sin \theta \hat{j}$

Final velocity = $u \cos \theta \hat{i}$

$$|\vec{v}_{\text{avg}}| = \frac{1}{2} \sqrt{(2u \cos \theta)^2 + (u \sin \theta)^2}$$

11. Ans (2)

$$H = H_1 + H_2$$

$$= \frac{200 \times 10^{-4}}{1} (60 - 20) + \frac{390 \times 10^{-4}}{1} (60 - 20)$$

$$= 0.8 + 1.56 = 2.36 \text{ J}$$

12. Ans (1)

Coefficient of volume expansion at constant pressure is $\frac{1}{273}$ for all gases. The average translational K.E. is same for molecules of all gases and for each molecule it is $\frac{3}{2} kT$

Mean free path $\lambda = \frac{kT}{\sqrt{2} \pi d^2 P}$ (as P decreases, λ increases)

13. Ans (2)

Mole remain constant

$$n = n_1 + n_2$$

$$\frac{P(2V)}{RT} = \frac{P_1 V}{R T_1} + \frac{P_2 V}{T_2}$$

$$\frac{P}{T} = \frac{1}{2} \left[\frac{P_1}{T_1} + \frac{P_2}{T_2} \right]$$

14. Ans (4)

Gravitational field on the surface of a solid sphere $I_g = \frac{GM}{R^2}$

By the graph

$$\frac{GM_1}{(1)^2} = 2$$

and $\frac{GM_2}{(2)^2} = 3$

On solving

$$\frac{M_1}{M_2} = \frac{1}{6}$$

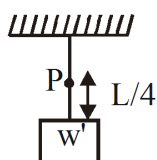
15. Ans (3)

By law of floating

$$Mg = \rho_1 V_1 g + \rho_2 V_2 g \Rightarrow M = \rho_1 V_1 + \rho_2 V_2$$

$$M = 0.6 \times 10^2 \times 4 + 0.4 \times 10^2 \times 6 = 480 \text{ gm}$$

Hence option (3)

16. Ans (2)


weight density, $\lambda = \frac{W}{L}$

weight of rope at length $\frac{L}{4} = \lambda \frac{L}{4}$

$$= \frac{W}{L} \times \frac{L}{4} = \frac{W}{4}$$

Total weight suspended from pt. P,

$$F = \frac{W}{4} + W'$$

$$\sigma = \frac{F}{A} = \frac{\left(\frac{W}{4} + W'\right)}{S}$$

17. Ans (1)

$$\frac{mv^2}{r} = \frac{k}{r^2} \Rightarrow mv^2 = \frac{k}{r} \therefore \text{K.E.} = \frac{1}{2} mv^2 = \frac{k}{2r}$$

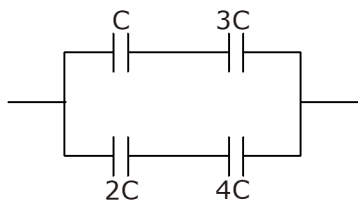
$$\text{P.E.} = \int F \, dr = \int \frac{k}{r^2} dr = -\frac{k}{r}$$

$$\therefore \text{Total energy} = \text{K.E.} + \text{P.E.} = \frac{k}{2r} - \frac{k}{r} = -\frac{k}{2r}$$

18. Ans (4)

$$E_{eq} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2} + \frac{E_3}{r_3}}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}}$$

19. Ans (2)

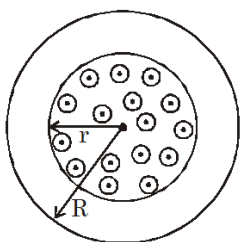


$$\text{Let } \frac{\epsilon_0 A}{2d} = C \quad \{\text{Area} = A/2\}$$

$$C_{\text{eq}} = \frac{3C^2}{C + 3C} + \frac{(2C)(4C)}{2C + 4C}$$

$$= \frac{3}{4}C + \frac{4}{3}C = \frac{25}{24} \frac{\epsilon_0 A}{d}$$

20. Ans (1)



$$E(2\pi R) = -\frac{d}{dt}[B(\pi r^2)]$$

$$\Rightarrow E = -\frac{r^2}{2R} \frac{dB}{dt}$$

$$\Rightarrow E dt = -\frac{r^2}{2R} dB$$

$$\Rightarrow \int qE dt = -\frac{qr^2}{2R} \int_B^0 dB$$

$$\Rightarrow mv = \frac{qr^2 B}{2R}$$

$$\Rightarrow v = \frac{qr^2 B}{2mR}$$

21. Ans (1)

$$\text{Volume of cube} \Rightarrow v = a^3$$

$$a = 1.2 \times 10^{-2} \text{ m}$$

Significant digit = 2

$$V = a^3 \Rightarrow V = (1.2 \times 10^{-2} \text{ m})^3$$

$$\Rightarrow V = 1.728 \times 10^{-6} \text{ m}^3$$

In this operation, minimum two significant digit is involved during multiplication of side.

The answer expressed with least significant digit hence volume expressed with two significant digit only.

$$V = 1.7 \times 10^{-6} \text{ m}^3$$

22. Ans (3)

Theoretical.

23. **Ans (1)**

$$\text{Speed of wave} = \frac{2 \times 10^{10}}{200} = 10^8 \text{ m/s}$$

$$\text{Refractive index} = \frac{3 \times 10^8}{10^8} = 3$$

$$\text{Now refractive index} = \sqrt{\epsilon_r \mu_r}$$

$$3 = \sqrt{\epsilon_r (1)}$$

$$\Rightarrow \epsilon_r = 9$$

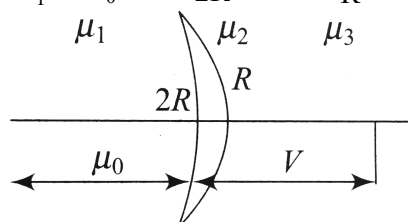
Option (1)

24. **Ans (2)**

$$\frac{\mu_2}{V} - \frac{\mu_1}{-u_0} = \frac{\mu_2 - \mu_1}{-2R}$$

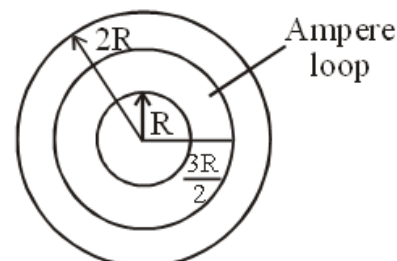
$$\frac{\mu_3}{V_f} - \frac{\mu_2}{V} = \frac{\mu_3 - \mu_2}{-R}$$

$$\frac{\mu_3}{V_f} + \frac{\mu_1}{u_0} = \frac{\mu_1 - \mu_2}{2R} + \frac{\mu_2 - \mu_3}{R}$$



$$\therefore \frac{\mu_1 - \mu_2}{2} + \mu_3 - \mu_2 \Rightarrow \mu_1 - \mu_2 < 2\mu_3 - 2\mu_2 \Rightarrow \mu_1 + \mu_2 < 2\mu_3$$

25. **Ans (3)**



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

$$B 2\pi \left(\frac{3R}{2} \right) = \frac{\mu_0 I \left(\pi \left(\frac{3R}{2} \right)^2 - \pi R^2 \right)}{\pi (2R)^2 - \pi R^2}$$

26. **Ans (2)**

$$\text{Energy } \frac{hc}{\lambda} = 4.5 \text{ eV}$$

Hence transition B

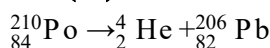
27. **Ans (4)**

$$\frac{1}{2} mv^2 = \frac{hc}{\lambda} - \phi$$

$$\frac{1}{2} mv'^2 = \frac{hc}{(3\lambda/4)} - \phi = \frac{4hc}{3\lambda} - \phi$$

$$\text{Clearly, } v' > \sqrt{\frac{4}{3}} v$$

28. **Ans (1)**



$$\text{mass defect} = \Delta m = (m_{\text{Po}} - m_{\text{He}} - m_{\text{Pb}})$$

$$= 0.005818\text{u}$$

$$\therefore Q = (\Delta m) \times 931.48\text{MeV} = 5419\text{keV}$$

From conservation of linear momentum, $P_{\text{b}} = P_{\alpha}$

$$\therefore \sqrt{2m_{\text{Pb}}K_{\text{Pb}}} = \sqrt{2m_{\alpha}K_{\alpha}} \Rightarrow \frac{K_{\alpha}}{K_{\text{Pb}}} = \frac{m_{\text{Pb}}}{m_{\alpha}} = \frac{206}{4}$$

$$\therefore K_{\alpha} = (Q) \left(\frac{206}{206 + 4} \right) \quad (K_{\text{total}} = Q)$$

$$= \frac{206}{210} \times 5419 \approx 5319\text{keV}$$

29. **Ans (2)**

$$f_{\text{before crossing}} = f_0 \left(\frac{c}{c - v_s} \right) = 1000 \left(\frac{320}{320 - 20} \right)$$

$$f_{\text{after crossing}} = f_0 \left(\frac{c}{c + v_s} \right) = 1000 \left(\frac{320}{320 + 20} \right)$$

$$\Delta f = f_0 \left(\frac{2cv_s}{c^2 - v_s^2} \right)$$

$$\frac{\Delta f}{f} \times 100\% = \frac{2 \times 320 \times 20}{300 \times 340} \times 100 = 12.54\% \approx 12\%$$

30. **Ans (1)**

I^{st} minima

$$\theta = 30$$

$$\sin \theta = \frac{n\lambda}{a}$$

$$\frac{1}{2} = \frac{1 \times \lambda}{1\mu\text{m}}$$

$$\lambda = 0.5\mu\text{m}$$

$$D = 0.5\text{ m}$$

$$\beta = 1 \times 10^{-2}\text{ m}$$

$$d = ?$$

$$\beta = \frac{D\lambda}{d}$$

$$10^{-2} = \frac{1 \times 0.5 \times 10^{-6}}{2d}$$

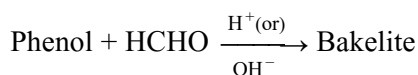
$$d = \frac{0.25 \times 10^{-6}}{10^{-2}}$$

$$d = 0.25 \times 10^{-4}\text{ m}$$

$$d = 25 \times 10^{-6}\text{ m}$$

$$d = 25\mu\text{m}$$

31. **Ans (3)**



32. **Ans (3)**

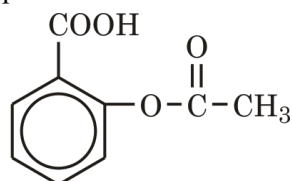
Vulcanization of rubber is carried out by heating a mixture of isoprene & sulphur

33. **Ans (1)**

It is used to separate liquid-liquid mixture which is immiscible with different densities

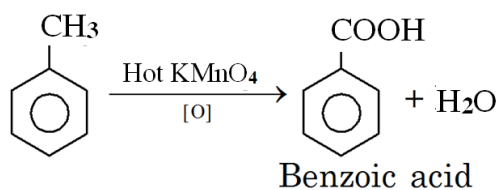
34. **Ans (1)**

Aspirin

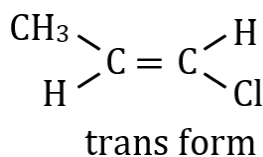
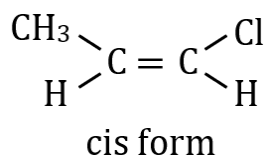


Acetyl salicylic acid

35. **Ans (1)**



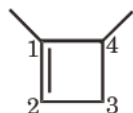
36. **Ans (2)**



37. **Ans (4)**

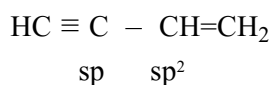
Tyrosine is not an essential amino acid.

38. **Ans (4)**



1,4-dimethyl butene

39. **Ans (3)**



40. **Ans (3)**

Reduction of Alkyl halide (except fluorides) produces alkane with zinc and dilute hydrochloric acids.

41. **Ans (3)**

$$\begin{aligned} \Delta_f H^\circ &= [\Delta_f H^\circ(\text{CO}_2, \text{g}) \times 1 + \Delta_f H^\circ(\text{H}_2\text{O}, \text{l}) \times 2] - [\Delta_f H^\circ(\text{CH}_4, \text{g})] \\ -890.4 &= -393.5 + (-285.9) \times 2 - \Delta_f H^\circ(\text{CH}_4, \text{g}) \\ \Delta_f H^\circ &= -395.5 - 571.8 + 890.4 = -74.9 \end{aligned}$$

42. **Ans (2)**

For Lyman series of He^{+2}

$$\frac{1}{\lambda} = R_H(2)^2 \left(\frac{1}{(1)^2} - \frac{9}{(\infty)^2} \right)$$

$$\frac{1}{\lambda} = R_H(2)^2$$

$$\lambda = \frac{1}{4 R_H}$$

For Balmer series of Li^{+2}

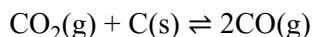
$$\frac{1}{\lambda} = R_H(3)^2 \left(\frac{1}{(2)^2} - \frac{1}{(\infty)^2} \right)$$

$$\frac{1}{\lambda} = \frac{9 R_H}{4}$$

Given that λ for Lyman of $\text{He}^+ = x \text{Å}$

$$\text{So } \lambda = \frac{4(4)}{9(4 R_H)} = \frac{16 x^0}{9} \text{Å}$$

43. **Ans (1)**



Initial 0.5 atm

At eqm. (0.5 - x) 2x

$$\therefore (0.5 - x) + 2x = 0.8 \text{ or } x = 0.3 \text{ atm}$$

$$\therefore P_{\text{CO}_2} = 0.5 - 0.3 = 0.2$$

$$P_{\text{CO}} = 2x = 2 \times 0.3 = 0.6 \text{ atm}$$

44. **Ans (2)**

$$\text{Average rate} = \frac{\text{Change in concentration}}{\text{Time}}$$

$$\text{The change in molarity} = \frac{\Delta n}{V} = \frac{\Delta P}{RT} = 0.0327$$

$$\begin{aligned} \text{Rate of reaction} &= \frac{0.0327}{50 \times 60} \\ &= 1.09 \times 10^{-5} \text{ mol/litre second} \end{aligned}$$

45. **Ans (3)**

$$500 \times 2 = 0.5 \times V$$

$$\Rightarrow V = 2000 \text{ ml}$$

$$\text{added volume} = 2000 - 500 = 1500 \text{ ml}$$

46. **Ans (1)**

$$\text{Cr}^{2+}|\text{Cr}^{3+} = +0.41\text{V}$$

$$\text{Mn}^{2+}|\text{Mn}^{3+} = -1.57\text{V}$$

$$\text{Fe}^{2+}|\text{Fe}^{3+} = -0.77\text{V}$$

$$\text{Co}^{2+}|\text{Co}^{3+} = -1.97\text{V}$$

As Cr will have maximum oxidation potential value, therefore its oxidation will be easiest.

47. **Ans (2)**

$$[H^+] = \frac{50 \times 10^{-1} + 50 \times 10^{-2}}{100}$$

$$= 5.5 \times 10^{-2}$$

$$\text{pH} = 2 - 0.74 = 1.26$$

48. **Ans (2)**

$$d_{\text{sample}} = 4 \text{ gm/lit.} = 4 \text{ kg/m}^3$$

$$V_{\text{rms}} = \sqrt{\frac{3P}{d}} = \sqrt{\frac{3 \times 1.2 \times 10^5}{4}} = 300 \text{ m/sec}$$

49. **Ans (2)**

the relative lowering of vapour pressure is equal to the mole fraction of solute

50. **Ans (2)**

$$C = \frac{40.7}{12} = 3.3916$$

$$H = \frac{5.1}{1} = 5.1$$

$$O = \frac{54.2}{16} = 3.387$$

Their simple ratio comes to be 2 : 3 : 2.

So the empirical formula = $C_2H_3O_2$.

51. **Ans (3)**

Total no. of atoms in fcc = 4

assuming atom to be spherical, its volume = $\frac{4}{3}\pi r^3$

Total volume of all atoms present in fcc = $\frac{16}{3}\pi r^3$

52. **Ans (2)**

$$T \propto \frac{1}{\sqrt{V}}$$

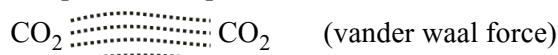
$$TV^{1/2} = \text{constant}$$

For adiabatic process, $TV^{1/2} = TV^{\gamma-1} = \text{constant}$

$$\therefore \gamma - 1 = \frac{1}{2}, \gamma = \frac{3}{2}$$

53. **Ans (4)**

Nonpolar + Nonpolar $\rightarrow$ London force



54. **Ans (3)**

$$[\text{Mn}(\text{CO})_6] : \text{EAN} = 37$$

55. **Ans (4)**

As chelation $\uparrow$, stability $\uparrow$.

56. **Ans (3)**

The incorrect order of increasing melting point is

$$\text{Ti} < \text{V} < \text{Cr} < \text{Mn}$$

Melting point increases from Ti to chromium but decreases in case of Mn. Mn has melting point even lower than that of Ti.

57. **Ans (1)**

The addition of phosphate fertilizers to water bodies causes enhanced growth of algae.

Hence option (a) is the answer

58. **Ans (4)**

Actinoid compounds are more basic than lanthanoid compounds.

59. **Ans (3)**

R_3SiCl is used to control chain length of silicone polymer.

$R \rightarrow$ Alkyl group $Me \rightarrow -CH_3$

60. **Ans (2)**

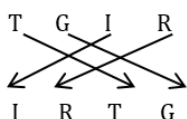
Cement contains elements of group 2 (Ca), group 13 (Al) and group 14 (Si).

71. **Ans (3)**

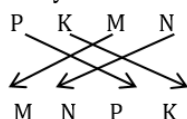
As, Admonish is less intense than berate similarly, defeat is less intense than conquer.

72. **Ans (2)**

As,



Similarly,

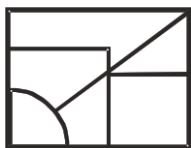


73. **Ans (3)**

Overlapping of two designs of first figure becomes the second figure.

74. **Ans (2)**

By observation,



75. **Ans (4)**

By observation,



76. **Ans (2)**

$$(10 \times 5) - (10 \div 2) = 45$$

$$(40 \times 5) - (40 \div 2) = 180$$

$$(60 \times 5) - (60 \div 2) = 270$$

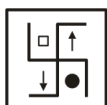
But same rule does not follow in option (2).

77. **Ans (2)**

All except spanner are tools used by gardener, while spanner is used by carpenter.

78. **Ans (4)**

By observation,



79. **Ans (4)**



80. **Ans (3)**

By observation,



81. **Ans (4)**

$$13 + (7 \times 2) = 27$$

$$54 + (45 \times 2) = 144$$

$$4 + (32 \times 2) = 68$$

82. **Ans (4)**

Row wise total sum = 30 i.e.

$$9 + 9 + 12 = 30$$

$$13 + 10 + 7 = 30$$

$$8 + 11 + 11 = 30$$

Hence, option (4) is correct.

83. **Ans (3)**

The given sequence 2, 17, 52,, 206 follow a pattern :

$$1^3 + 1^2 = 2,$$

$$2^3 + 3^2 = 17,$$

$$3^3 + 5^2 = 52,$$

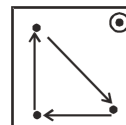
$$4^3 + 7^2 = 113,$$

$$5^3 + 9^2 = 206.$$

ence, 113 is the missing number.

84. **Ans (2)**

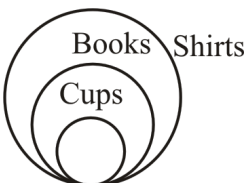
In the first step, the elements at the four ends move in the sequence



. In each subsequent step, the

elements move in the sequence obtained by rotating the previous sequences through 90° ACW. Also, in each step, the element that reaches the encircled position gets replaced by a new element.

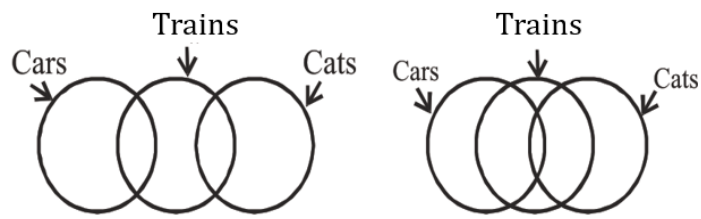
85. **Ans (2)**



Only conclusion 2 follows.

86. **Ans (3)**

For the given statements two possible cases are possible.



87. **Ans (1)**

By observation,



88. **Ans (1)**

By observation,



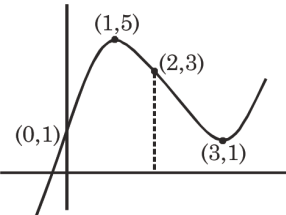
89. **Ans (1)**

It is mentioned in the passage that though India boasts of huge technical manpower, yet it lags behind other countries in technology. Thus, it is quite probable that India is in no way at par with the developed countries in terms of technological development.

90. **Ans (3)**

According to the passage, the compensation package in different companies includes different provisions.

91. **Ans (1)**



$$\therefore M = 5 \text{ and } m = 1$$

$$M - 2m = 3$$

92. **Ans (1)**

$$x^3 - y^2 = 0 \quad \dots(1)$$

$$2y \frac{dy}{dx} = 3x^2$$

$\therefore$ Slope of tangent at

$$P = \left. \frac{dy}{dx} \right|_{(4m^2, 8m^3)} = \left(\frac{3x^2}{2y} \right)_{(4m^2, 8m^3)} = 3m$$

$\therefore$ Equation of tangent at P is

$$y - 8m^3 = 3m(x - 4m^2)$$

$$\text{or } y = 3mx - 4m^3 \quad \dots(2)$$

It cuts the curve again at Q,

From (1) & (2) ;

$$x = 4m^2, m^2$$

Put $x = 8m^2$ in (2) ; then

$$y = -m^3$$

$$\therefore Q(m^2, -m^3)$$

Slope of tangent at

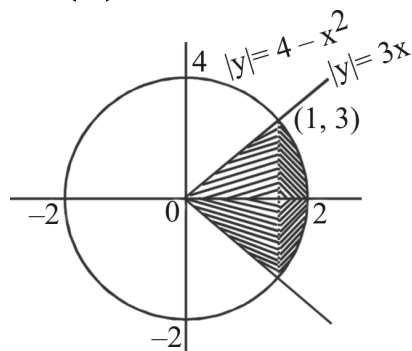
$$Q = \left(\frac{dy}{dx} \right)_{(m^2, -m^3)} = -\frac{3}{2}m$$

$$\text{Slope of Normal at } Q = \frac{2}{3}m$$

$$\therefore \frac{2}{3m} = 3m$$

$$\text{or } 9m^2 = 2$$

93. **Ans (2)**



$$\text{Area} = 2 \left[\frac{3}{2} + \int_1^2 (4 - x^2) dx \right]$$

$$= \frac{19}{3} = 6 + \frac{1}{3}$$

$$\therefore K = 2$$

94. **Ans (2)**

$$\text{Now, } 1 + x = \log_a a + \log_a bc = \log_a abc$$

$$\Rightarrow \frac{1}{1+x} = \log_{abc} a$$

$$\text{Similarly, } \frac{1}{1+y} = \log_{abc} b \text{ and } \frac{1}{1+z} = \log_{abc} c$$

$$\therefore \frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z}$$

$$= \log_{abc} a + \log_{abc} b + \log_{abc} c$$

$$= \log_{abc} abc = 1$$

95. **Ans (3)**

$$x^2 + x + 1 > 0 \quad \forall x \in \mathbb{R}$$

$$\text{so } |x^2 - 3x - 1| < 3(x^2 + x + 1)$$

$$\Rightarrow (x^2 - 3x - 1)^2 - [3(x^2 + x + 1)]^2 < 0$$

$$\Rightarrow (4x^2 + 2)(-2x^2 - 6x - 4) < 0$$

$$\Rightarrow (x+2)(x+1) > 0 \Rightarrow x \in (-\infty, -2) \cup (-1, \infty)$$

96. **Ans (2)**

$$\text{Given sum is write as } \sum_{r=0}^n \frac{C_r^2}{r+1}$$

$$= \frac{1}{n+1} \sum \left(\frac{n+1}{r+1} {}^nC_r \right) ({}^nC_r)$$

$$= \frac{1}{n+1} \sum_{r=0}^n ({}^{n+1}C_{r+1}) ({}^nC_r)$$

$$= \frac{1}{n+1} [{}^{n+1}C_1 {}^nC_0 + {}^{n+1}C_2 {}^nC_1 + \dots + {}^{n+1}C_{n+1} {}^nC_n]$$

$$= \frac{1}{n+1} [\text{coefficient of } x_{n+1} \text{ product of } (1+x)^{n+1} (x+1)^n]$$

$$= \frac{1}{n+1} [{}^{2n+1}C_{n+1}] = \frac{1}{n+1} {}^{2n+1}C_n$$

97. **Ans (1)**

$$f(x) = \begin{cases} a|\pi - x| + 1; x \geq 5 \\ b|\pi - x| + 3; x < 5 \end{cases}$$

$$a|\pi - 5| + 1 = b|5 - \pi| + 3$$

$$a(5 - \pi) + 1 = b(5 - \pi) + 3$$

$$(a - b)(5 - \pi) = 2$$

$$a - b = \frac{2}{5 - \pi}$$

98. **Ans (1)**

$$|2z-1| = |z+2i|$$

$$\Rightarrow |(2x-1) + (2iy)| = |x + i(y+2)|$$

$$\Rightarrow 3x^2 + 3y^2 - 4x - 4y - 3 = 0 \text{ i.e. Circle}$$

99. Ans (2)

The given equation is

$$|z|^n = (z^2 + z) |z|^{n-2} + 1$$

$$\Rightarrow z^2 + z \text{ is real}$$

$$\Rightarrow z^2 + z = \bar{z}^2 + \bar{z}$$

$$\Rightarrow (z - \bar{z})(z + \bar{z} + 1) = 0$$

$$\Rightarrow z = \bar{z} = x \text{ as } z + \bar{z} + 1 \neq 0 \text{ (} x \neq -1/2 \text{)}$$

Hence, the given equation reduces to

$$x^n = x^n + x|x|^{n-2} + 1$$

$$\Rightarrow x|x|^{n-2} = -1$$

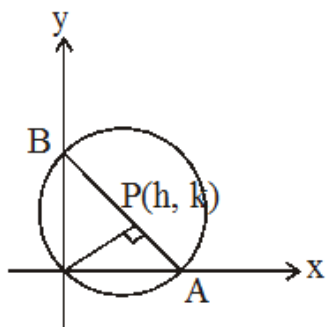
$$\Rightarrow x = -1$$

So number of solution is 1.

100. Ans (3)

$$\text{Slope of AB} = \frac{-h}{k}$$

$$\text{Equation of AB is } hx + ky = h^2 + k^2$$



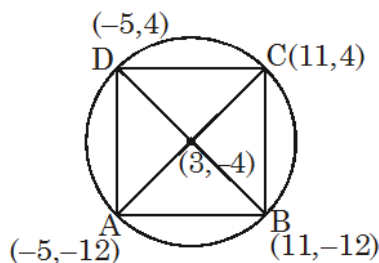
$$A \left(\frac{h^2 + k^2}{h}, 0 \right), B \left(0, \frac{h^2 + k^2}{k} \right)$$

$$AB = 2R$$

$$\Rightarrow (h^2 + k^2)^3 = 4R^2 h^2 k^2$$

$$\Rightarrow (x^2 + y^2)^3 = 4R^2 x^2 y^2$$

101. Ans (4)



$$R = \sqrt{9 + 16 + 103} = 8\sqrt{2}$$

$$OA = 13$$

$$OB = \sqrt{265}$$

$$OC = \sqrt{137}$$

$$OD = \sqrt{41}$$

102. Ans (2)

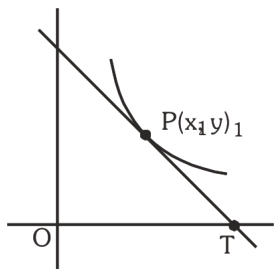
Re-arranging the equation ; we have

$$dx - y dy + \sqrt{x^2 + y^2} (x dx + y dy) = 0$$

$$\Rightarrow dx - y dy + \frac{1}{2} \sqrt{x^2 + y^2} d(x^2 + y^2) = 0$$

$$\Rightarrow x - \frac{y^2}{2} + \frac{1}{3} (x^2 + y^2)^{3/2} = C$$

103. Ans (1)



Equation of tangent $y - y_1 = m(x - x_1)$

Coordinate of T $\left(x - \frac{dx}{dy} y, 0 \right)$

Given $x - \frac{dx}{dy} y = ky$

$$\Rightarrow \frac{dx}{dy} + \left(-\frac{1}{y} \right) x = -k \text{ . Its linear}$$

$$\text{IF} = e^{-\log y} = \frac{1}{y}$$

$$\text{Solution } x \cdot \frac{1}{y} = \int -\frac{k}{y} dx \Rightarrow \frac{x}{y} = -k \log y + c$$

$$\Rightarrow y = ae^{-x/ky}$$

104. Ans (1)

$$y = \tan^{-1} \sqrt{\left(\frac{1 - \cos\left(\frac{\pi}{2} + x\right)}{1 + \cos\left(\frac{\pi}{2} + x\right)} \right)}$$

$$= \tan^{-1} \left| \tan\left(\frac{\pi}{4} + \frac{x}{2}\right) \right| \dots\dots(i)$$

Now, $\frac{\pi}{2} < x < \pi$

$$\therefore \frac{\pi}{4} < \frac{x}{2} < \frac{\pi}{2}$$

$$\text{or } \frac{\pi}{2} < \frac{\pi}{4} + \frac{x}{2} < \frac{3\pi}{4}$$

$$\therefore \left| \tan\left(\frac{\pi}{4} + \frac{x}{2}\right) \right| = -\tan\left(\frac{x}{4} + \frac{x}{2}\right)$$

($\because$ in II quadrant)

$$= \tan\left\{ \pi - \left(\frac{\pi}{4} + \frac{x}{2}\right) \right\}$$

From Eq.(i)

$$y = \tan^{-1} \tan\left\{ \pi - \left(\frac{\pi}{4} + \frac{x}{2}\right) \right\}$$

$$= \pi - \left(\frac{\pi}{4} + \frac{x}{2}\right)$$

$$= \frac{3\pi}{4} - \frac{x}{2}$$

($\because$ principal value of $\tan^{-1} \tan x$ in $-\frac{\pi}{2}$ to $\frac{\pi}{2}$)

$$\therefore \frac{dy}{dx} = -\frac{1}{2}$$

105. Ans (3)

$$I = \int_0^{\pi/4} f(x) dx + \int_{\pi/4}^{\pi/2} f(x) dx$$

where $f(x) = |\sin x - \cos x|$

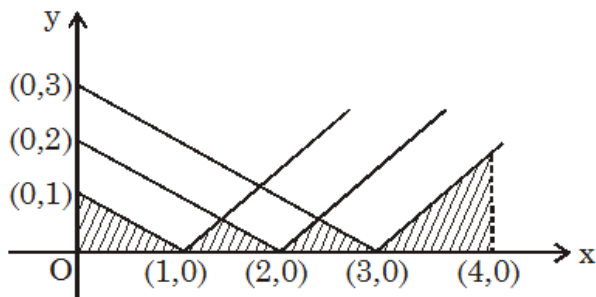
$$= \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx$$

$$= [\sin x + \cos x]_0^{\pi/4} + [-\cos x - \sin x]_{\pi/4}^{\pi/2}$$

$$= 2\left(\frac{1}{\sqrt{2}}\right) - 1 + (-1) + 2\left(\frac{1}{\sqrt{2}}\right)$$

$$= 2(\sqrt{2} - 1)$$

106. Ans (2)



Required value

$$= \frac{1}{2}(1)(1) + 2\left(\frac{1}{2} \times 1 \times \frac{1}{2}\right) + \frac{1}{2} \times 1 \times 1 = \frac{3}{2}$$

107. Ans (1)

$$\Delta = x^k y^k z^k \begin{vmatrix} 1 & ar & a^2 r^2 \\ 1 & ar^2 & a^2 r^4 \\ 1 & ar^3 & a^2 r^6 \end{vmatrix} = a^{3k} \cdot r^{6k} \cdot a^3 r^3 \begin{vmatrix} 1 & 1 & 1 \\ 1 & r & r^2 \\ 1 & r^2 & r^4 \end{vmatrix}$$

$$= a^{3(k+1)} \cdot r^{6k+3} (1-r)(r-r^2)(r^2-1)$$

Clearly, $k = -1$

$$\therefore \Delta = r^{-2} (1-r)^2 (r^2-1) = (r-1)^2 \left(1 - \frac{1}{r^2}\right)$$

108. Ans (2)

The radius of the circle having SS' as diameter is $r = ae$.

If it cuts an ellipse, then $r > b$ or $ae > b$ or $e^2 > \frac{b^2}{a^2}$ or $e^2 > 1 - e^2$

$$\text{or } e^2 > \frac{1}{2} \text{ or } e > \frac{1}{\sqrt{2}} \text{ or } e \in \left(\frac{1}{\sqrt{2}}, 1\right)$$

109. Ans (3)

Let ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\therefore$ It passes through $(4, -1)$

$$\Rightarrow \frac{16}{a^2} + \frac{1}{b^2} = 1 \quad \dots(1)$$

$\therefore y = -\frac{1}{4}x + \frac{5}{2}$ is a tangent to it

$$\text{hence } \sqrt{\frac{a^2}{16} + b^2} = \frac{5}{2} \quad \dots(2)$$

Solving (1) & (2) gives $a^2 = 20$ and $b^2 = 5$

110. Ans (2)

$$f(x) = \log_2 [3x - 3[x]]$$

$$f(x) = \log_2 [3\{x\}]$$

$$3\{x\} \in [0, 3)$$

$$\& [3\{x\}] = 0, 1, 2$$

$$\therefore \text{Range of } f(x) = 0, 1$$

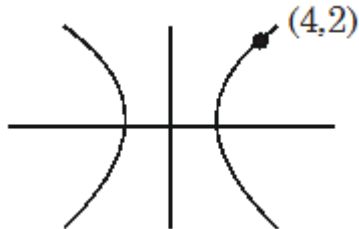
111. Ans (1)

$x + \sin x$ is strictly increasing function

if $x + \sin x$ then $e^{x+\sin x}$ also $\Rightarrow f(x)$ is one-one

$e^{x+\sin x} \in (0, \infty) \Rightarrow f(x)$ is into function

112. Ans (1)



$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$2a = 4 \Rightarrow a = 2$$

$$\frac{x^2}{4} - \frac{y^2}{b^2} = 1$$

Passes through $(4, 2)$

$$4 - \frac{4}{b^2} = 1 \Rightarrow b^2 = \frac{4}{3} \Rightarrow e = \frac{2}{\sqrt{3}}$$

113. Ans (3)

$$\text{Put } \sin x = \left(\frac{2 \tan x/2}{1 + \tan^2 x/2} \right)$$

$$\& \cos x = \left(\frac{1 - \tan^2 x/2}{1 + \tan^2 x/2} \right)$$

114. **Ans (3)**

$$\sin^{-1} \theta = \sin^{-1} (\sin 5)$$

$$\sin^{-1} \theta = 5 - 2\pi$$

$$\theta = \sin (5 - 2\pi)$$

115. **Ans (3)**

$$\begin{aligned} & \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x (1 - \sin x)}{8 \left(\frac{\pi}{2} - x \right)^3} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin \left(\frac{\pi}{2} - x \right) \left(1 - \cos \left(\frac{\pi}{2} - x \right) \right)}{8 \times \left(\frac{\pi}{2} - x \right) \times \left(\frac{\pi}{2} - x \right)^2} \\ &= \frac{1}{8} \times 1 \times \frac{1}{2} \\ &= \frac{1}{16} \end{aligned}$$

116. **Ans (4)**

$$\begin{aligned} |2A^9 B^{-1}| &= \frac{2^2 |A|^9}{|B|} \\ &= \frac{4 \times (-1)^9}{2} = -2 \end{aligned}$$

117. **Ans (2)**

$$P = \frac{{}^6C_3}{6^3} = \frac{20}{216}$$

Fav Case = 6C_3 Three numbers can be selected from six number & there is only one way in which three selected number can appears.

118. **Ans (2)**

Two balls transferred from I to II can be 1W and 1B or both W or both B.

Then the required probability

$$\begin{aligned} &= \left(\frac{{}^4C_1 {}^3C_1}{{}^7C_2} \right) \cdot \frac{6}{9} + \left(\frac{{}^4C_2}{{}^7C_2} \right) \cdot \frac{7}{9} + \left(\frac{{}^3C_2}{{}^7C_2} \right) \cdot \frac{5}{9} \\ &= \frac{43}{63} = \frac{p}{q} \therefore p + q = 106 \end{aligned}$$

119. **Ans (2)**

Considering CC as single object we will arrange CC, U, E at three places = 3!

$$\times CC \times U \times E \times$$

At thrse four vacant places 3 S can be arranged by $\frac{{}^4P_3}{3!}$

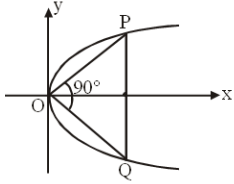
so required number of words = $3! \times \frac{{}^4P_3}{3!} = 24$

120. **Ans (2)**

If $P(at_1^2, 2at_1)$ be one end of the normal, the other say $Q(at_2^2, 2at_2)$

$$\text{then } t_2 = -t_1 - \frac{2}{t_1} \quad \dots (1)$$

$$\text{Again slope of } OP = \frac{2at_1}{at_1^2} = \frac{2}{t_1}$$



$$\text{Slope of } OQ = \frac{2}{t_2} \quad \therefore \frac{2}{t_1} \times \frac{2}{t_2} = -1$$

$$\Rightarrow t_1 t_2 = -4 \quad \dots (2)$$

from (1) and (2)

$$-\frac{4}{t_1} = -t_1 - \frac{2}{t_1} \Rightarrow +\frac{2}{t_1} = t_1 \Rightarrow t_1^2 = 2$$

$$\Rightarrow t_1 = \pm \sqrt{2}$$

Note : Slope of the normal at P is $(-t_1)$

121. **Ans (3)**

After reflector ray must pass through focus.

S (0,2)

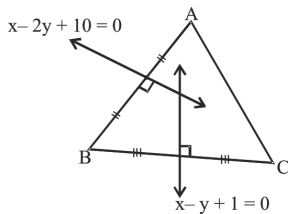
122. **Ans (1)**

$$A > 0, B > 0, C < 0$$

So both roots of equation are negative and $\alpha < \beta$ (given).

$$\text{So } \beta < \alpha < 0$$

123. **Ans (2)**



Circumcenter is the point of intersection of the perpendicular bisectors of the sides.

124. **Ans (1)**

In 17, 21, 25,, 417, number of terms = 101.

$$\text{and } T_n = 17 + (n - 1)4.$$

In 16, 21, 26,, 466, number of terms = 91

$$\text{and } T_m = 16 + (m - 1)5.$$

$$T_n = T_m \Rightarrow 17 + (n - 1)4 = 16 + (m - 1)5, 1 < n < 101, \\ \text{and } 1 < m < 91$$

$$\therefore 5m - 4n - 2 = 0 \Rightarrow m = 2$$

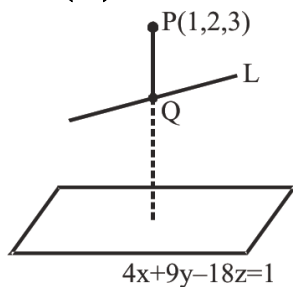
$$\therefore 5m - 4n - 2 = 0 \Rightarrow m = 2 \text{ is a solution.}$$

$$\text{Then } n = 2, 7, \dots,; 2 + (k - 1)5 < 101 \Rightarrow k = 20.$$

Also $m = 2, 6, 10, \dots$

Hence, there are 20 common terms.

125. Ans (4)



Any point on line is

$$Q(1-2\lambda, -1+3\lambda, 5+4\lambda)$$

$\therefore$ Direction ratios of PQ

are $(-2\lambda, 3\lambda-3, 4\lambda+2)$

PQ is perpendicular to the plane

$\Rightarrow$ PQ is parallel to normal of the plane

$$\Rightarrow \frac{-2\lambda}{4} = \frac{3\lambda-3}{9} = \frac{4\lambda+2}{-18}$$

$$\Rightarrow \lambda = \frac{2}{5}$$

126. Ans (1)

perpendicular vector to the plane

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ -1 & 1 & -2 \end{vmatrix} = -3\hat{i} + 3\hat{j} + 3\hat{k}$$

Eq. of plane

$$-3(x-1) + 3(y-1) + 3z = 0$$

$$\Rightarrow x - y - z = 0$$

$$d_{(2,1,4)} = \frac{|2-1-4|}{\sqrt{1^2+1^2+1^2}} = \sqrt{3}$$

127. Ans (3)

$$16^{\sin^2 x} + 16^{1-\sin^2 x} = 10$$

$$\text{Put } 16^{\sin^2 x} = t$$

$$t + \frac{16}{t} = 10$$

$$t^2 - 10t + 16 = 0$$

$$\therefore t = 2 \quad \text{or} \quad t = 8$$

$$t = 2 \qquad t = 8$$

$$16^{\sin^2 x} = 2^1 \qquad 16^{\sin^2 x} = 2^3$$

$$\therefore \sin^2 x = \frac{1}{4} \qquad \sin^2 x = \frac{3}{4}$$

$$\sin x = \pm \frac{1}{2} \qquad \sin x = \pm \frac{\sqrt{3}}{2}$$

$$[0, 2\pi] \text{ total solution} = 8$$

128. Ans (3)

$$\begin{aligned}
 & \sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{5\pi}{8} + \sin^4 \frac{7\pi}{8} \\
 &= \frac{1}{4} \left[\left(2\sin^2 \frac{\pi}{8} \right)^2 + \left(2\sin^2 \frac{3\pi}{8} \right)^2 \right] \\
 &+ \frac{1}{4} \left[\left(2\sin^2 \frac{\pi}{8} \right)^2 + \left(2\sin^2 \frac{3\pi}{8} \right)^2 \right] \\
 &= \frac{1}{4} \left[\left(1 - \cos \frac{\pi}{4} \right)^2 + \left(1 - \cos \frac{3\pi}{4} \right)^2 \right] \\
 &+ \frac{1}{4} \left[\left(1 - \cos \frac{\pi}{4} \right)^2 + \left(1 - \cos \frac{3\pi}{4} \right)^2 \right] \\
 &= \frac{1}{4} \left[\left(1 - \frac{1}{\sqrt{2}} \right)^2 + \left(1 + \frac{1}{\sqrt{2}} \right)^2 \right] + \frac{1}{4} \left[\left(1 - \frac{1}{\sqrt{2}} \right)^2 + \left(1 + \frac{1}{\sqrt{2}} \right)^2 \right] \\
 &= \frac{1}{4}(3) + \frac{1}{4}(3) = \frac{3}{2}
 \end{aligned}$$

129. Ans (1)

$$\begin{aligned}
 \hat{a} + \hat{b} &= \left(\frac{7\hat{i} - 4\hat{j} - 4\hat{k}}{9} \right) + \left(\frac{2\hat{i} - \hat{j} + 2\hat{k}}{3} \right) \\
 &= \frac{(7\hat{i} - 4\hat{j} - 4\hat{k}) + (-6\hat{i} - 3\hat{j} - 6\hat{k})}{9} \\
 \therefore \text{Vector mag. } 3\sqrt{6} &= \frac{3\sqrt{6}(\hat{i} - 7\hat{j} + 2\hat{k})}{\sqrt{54}}
 \end{aligned}$$

130. Ans (3)

$$\begin{aligned}
 (\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) &= 2(\vec{b} \times \vec{a}) \\
 &= 2 \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -2 \\ 3 & 2 & 2 \end{vmatrix} = 2(8\hat{i} - 8\hat{j} + 4\hat{k}) \\
 \text{Required vector} &= \pm 12 \frac{(2\hat{i} - 2\hat{j} - \hat{k})}{3} \\
 &= \pm 4(2\hat{i} - 2\hat{j} - \hat{k})
 \end{aligned}$$